



Basic Principles of Medicine 1
Module: Musculoskeletal System | Lecture 05

Clinical Anatomy of the Arm & Forearm

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Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 4th ed. Elsevier Churchill Livingstone.
- Printed Textbook: Chapter 7: pg 753 - 782
- Online Textbook: Chapter 7: [click here](#)
 - Conceptual Overview
 - Regional Anatomy:
 - Posterior scapular region
 - Arm
 - Elbow joint
 - Forearm (anterior and posterior compartment)
 - Cubital fossa
- Pay special attention to the **green (printed text)** / **blue (ebook)** “*In the Clinic*” boxes for the clinical correlations for this topic

Objectives

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Describe cubital tunnel syndrome and the clinical consequences (SG Article)

List the general attachments, innervations, blood supply and actions of the muscles in the flexor and extensor compartments of the arm

Describe the cutaneous innervation of the arm

Describe the course of the brachial artery and its branches.

Identify the bones of the arm in imaging studies including and use this information to interpret clinical signs and symptoms.

Describe the movements that occur at the elbow joint

Describe the bony and ligamentous anatomy of the elbow joint.

Describe the movements that occur at the proximal and distal radioulnar joints

Describe the arterial anastomosis of the elbow.

Describe the borders of the cubital fossa, its contents and their relationships.

Describe the anatomical location, mechanism and clinical sequences of elbow joint dislocation

Describe the cause and signs of Volkmann's ischemic contracture.

Describe the basic mechanism of radial head subluxation ("Pulled elbow") in small children and identify it on a radiograph

Describe and identify fracture of head of radius and possible clinical consequences.

Describe the common flexor and common extensor tendons and the signs and symptoms of inflammation of these tendons

List the general attachments, innervations, blood supply and actions of the muscles in the flexor and extensor compartments of the forearm

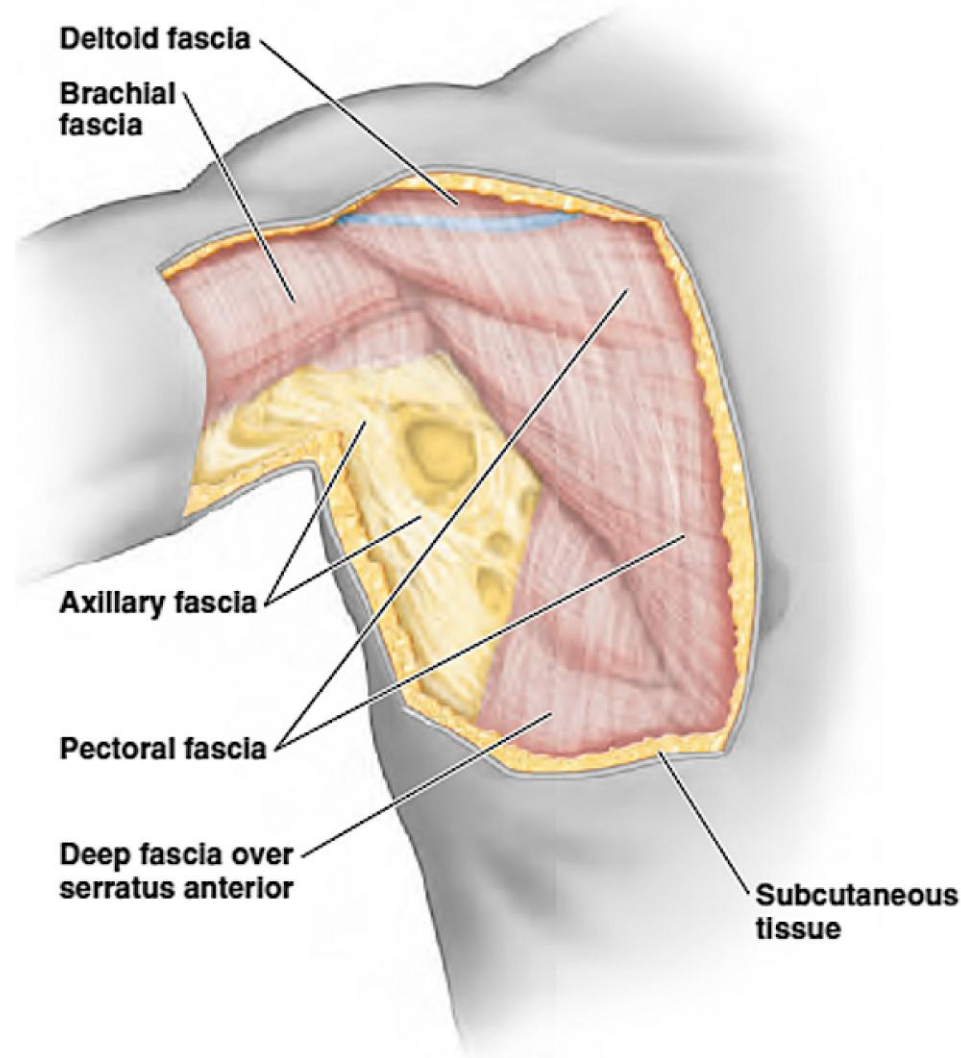
Describe the cutaneous innervation of the forearm

Describe the course of the radial and ulnar arteries in the forearm

Describe the landmarks used when taking the radial pulse.

Limb Compartments: General

- Region in which components share similar functions, innervation and blood supply.
- Formed by fascia
- **Deep fascia**
 - Forms an enveloping sheath
 - Lies deep to the superficial fascia (subcutaneous tissue), covering most of the body

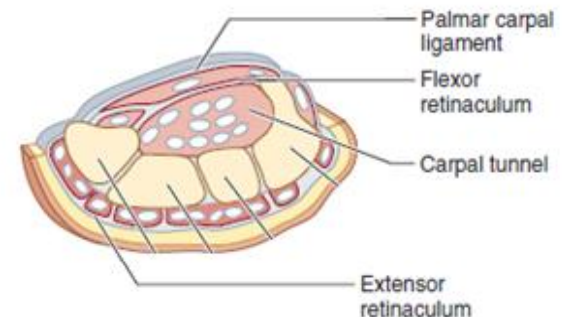
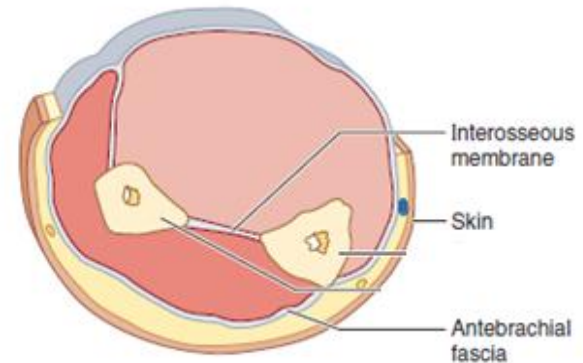
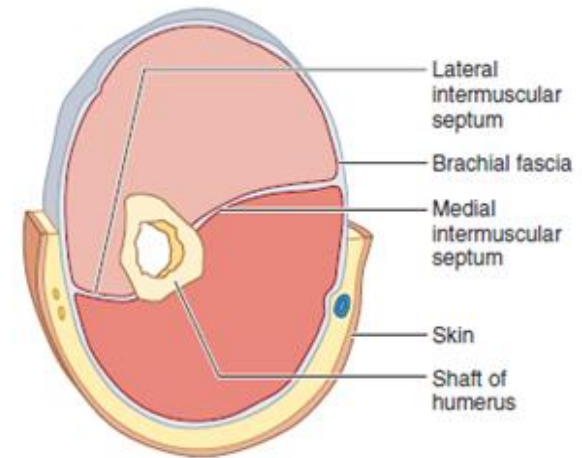


Upper Limb Compartments

- Compartments can be further subdivided by septa
- Intermuscular septa
 - Sheets of connective tissue from the bones to the deep fascia
 - If no structure intervenes between skin and bone, deep fascia usually attaches to bone

In the upper limb:

- **Brachial (arm) fascia**
 - continuous with fascia of shoulder/pectoral regions
- **Antebrachial (forearm) fascia**
- **Deep fascia of the hand**



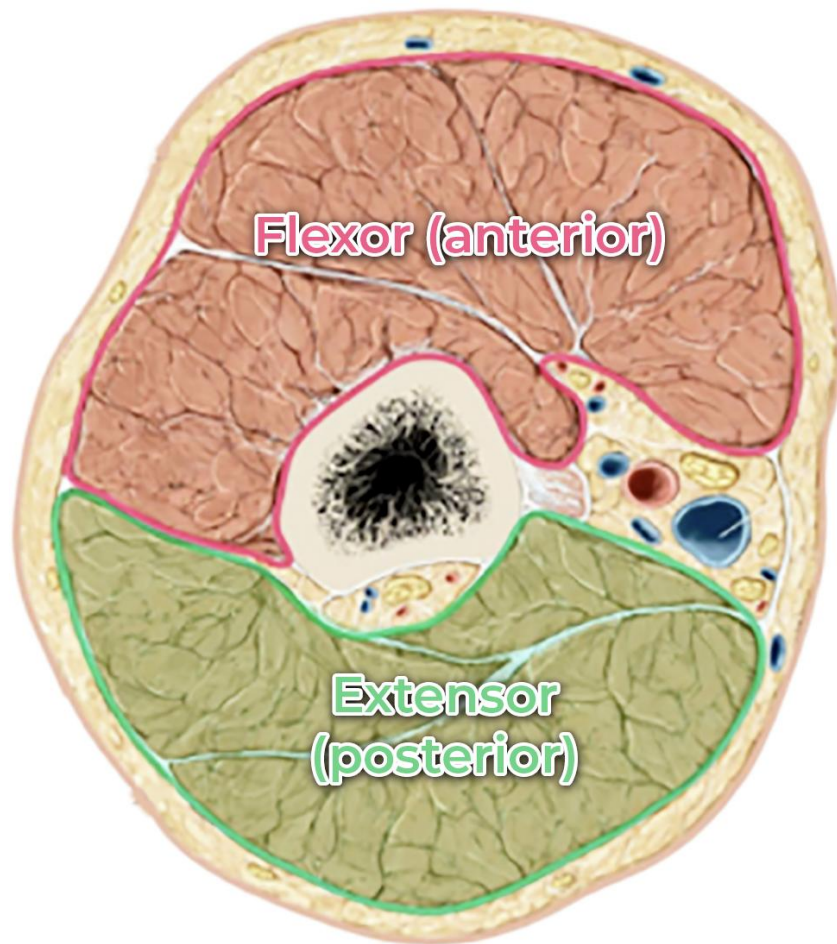
General Rule

- If a muscle crosses a joint, it will act on that joint

In the upper limb:

- If a muscle crosses anterior to a joint, it will **flex** that joint
- If a muscle crosses posterior to a joint, it will **extend** that joint

Anterior



Posterior

Arm Compartments: Overview

- Anterior (flexor) compartment
 - Musculocutaneous nerve
- Posterior (extensor) compartment
 - Radial nerve

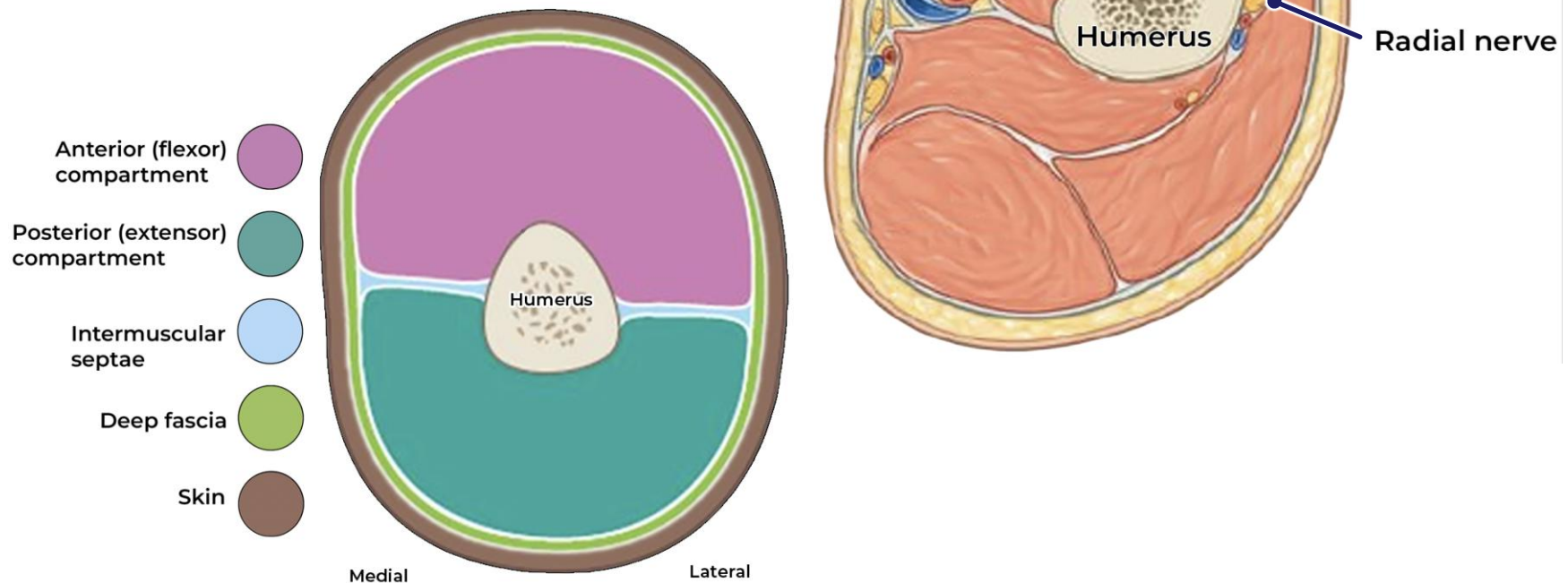
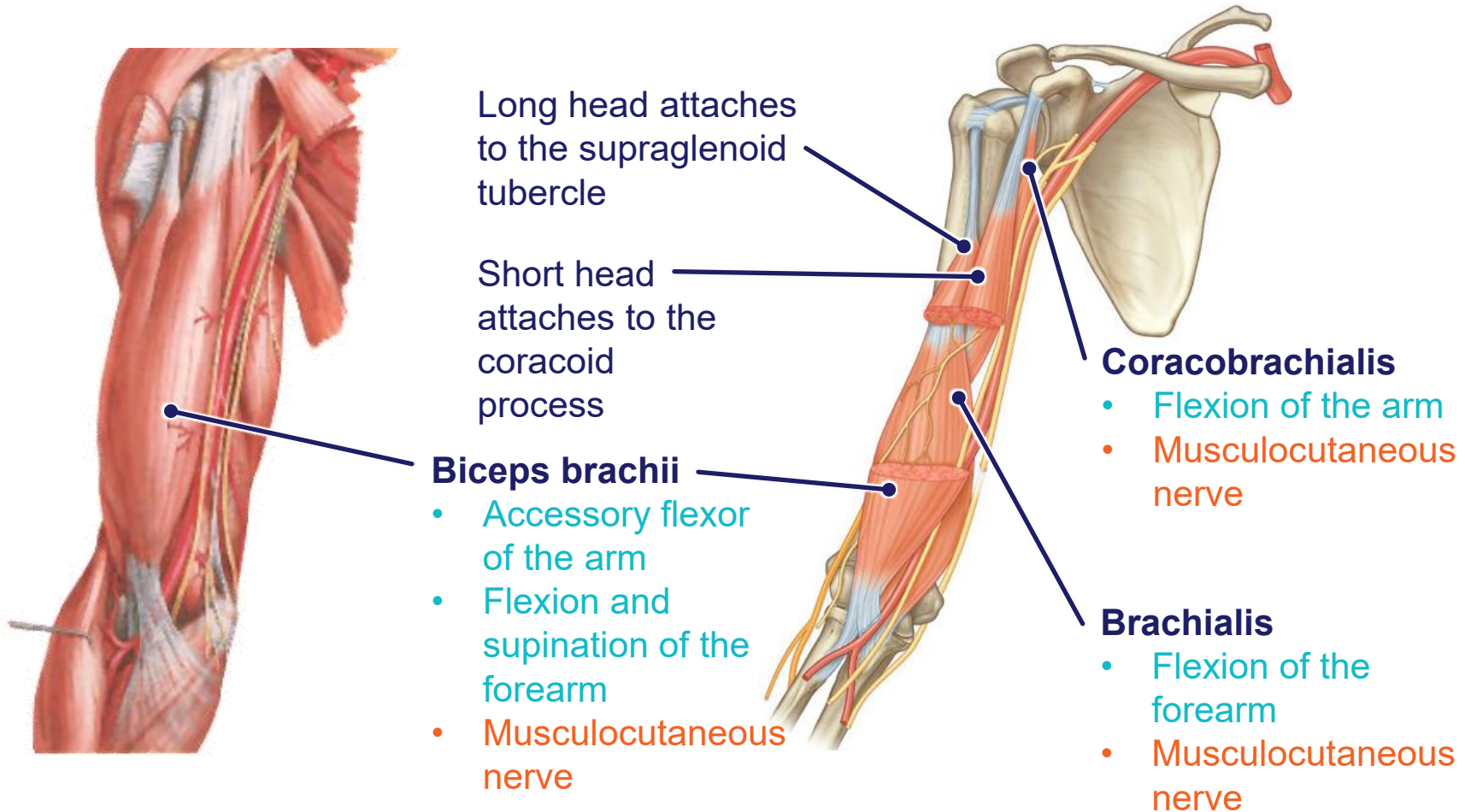
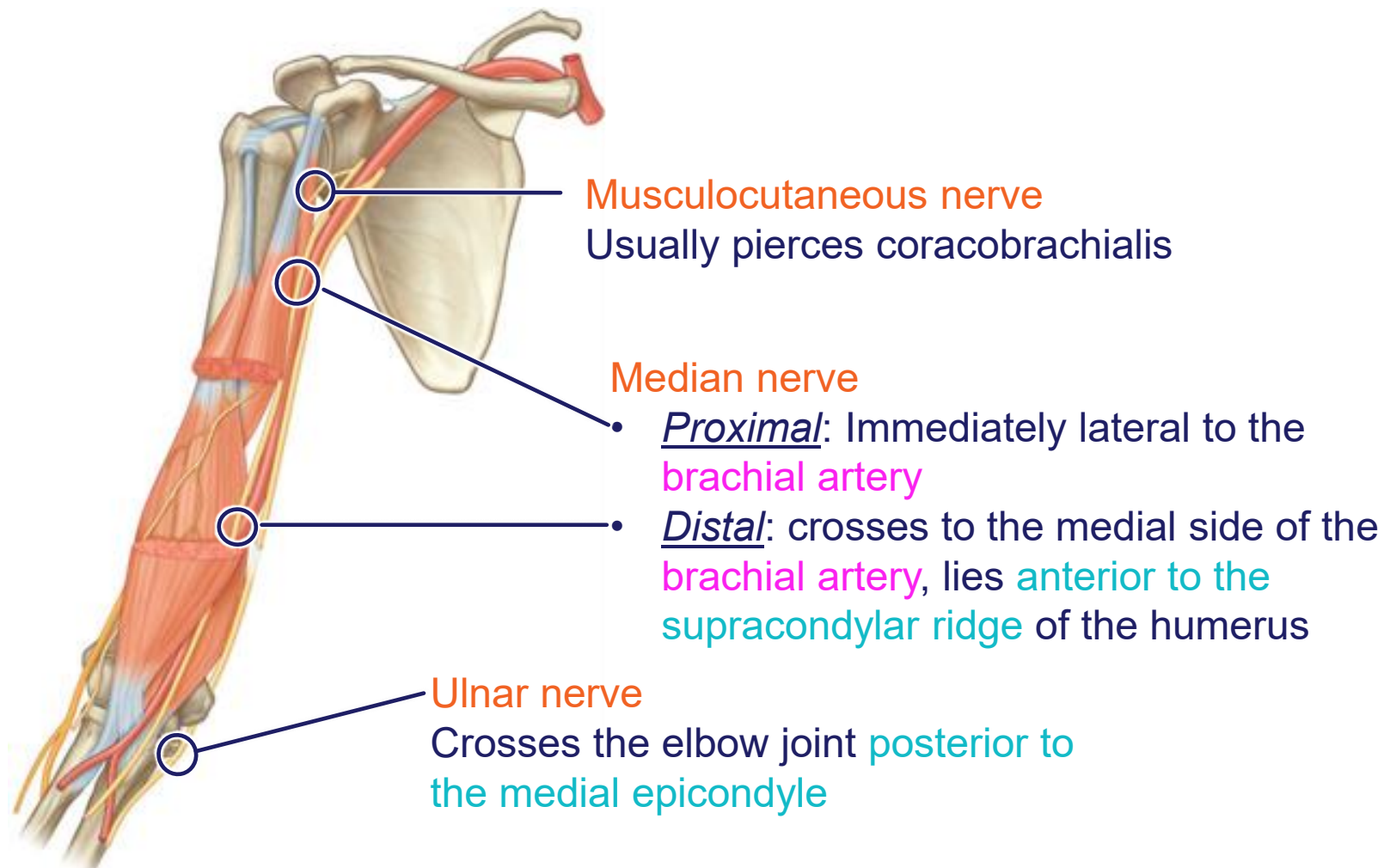


Fig. 15.25

Anterior Compartment



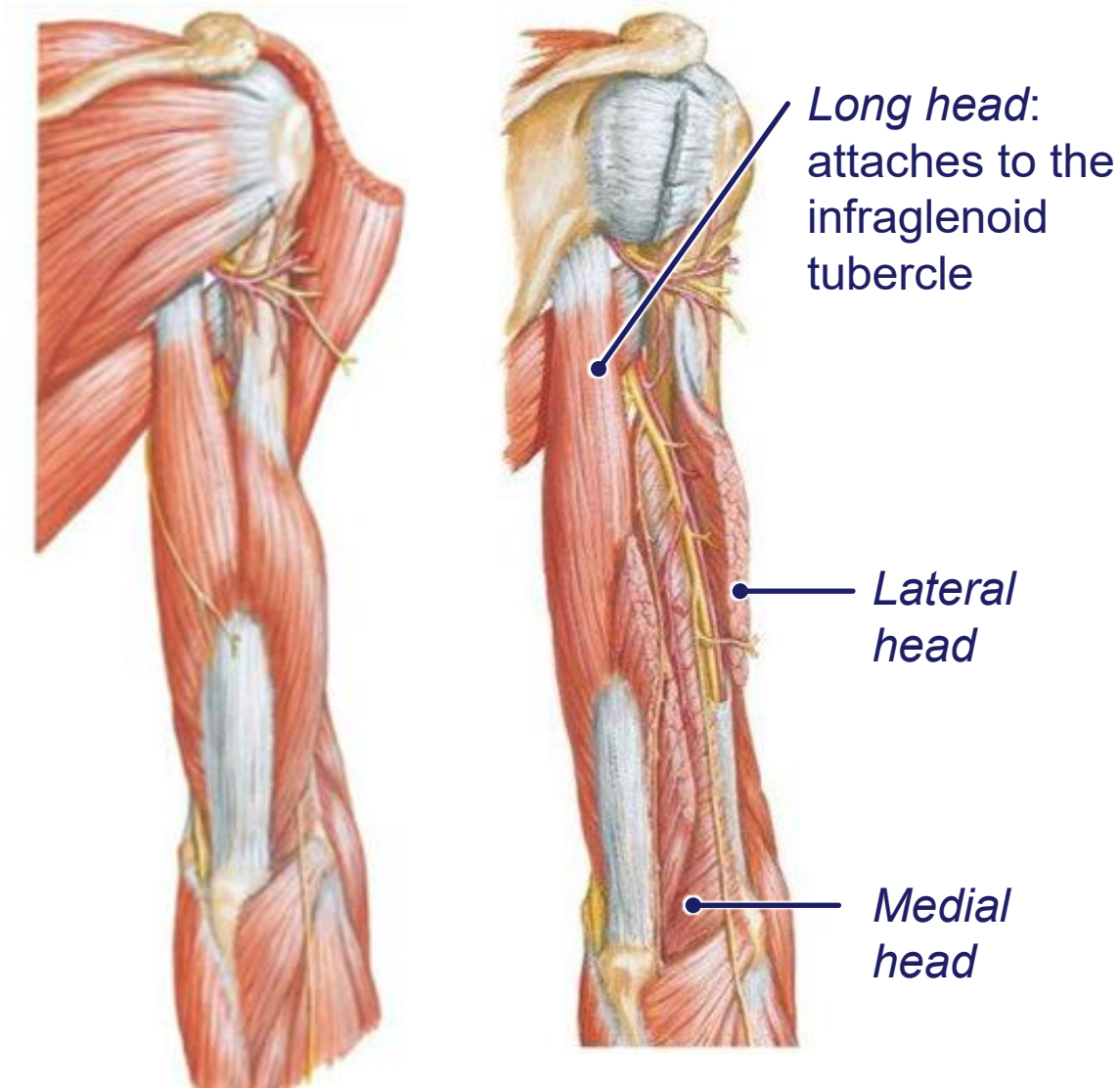
Neurovascular courses in the Anterior Arm



Posterior Compartment

Triceps brachii

- Extension of the forearm
- Assists in adduction of the arm (through long head)
- Radial nerve



Neurovascular courses in the Posterior Arm



Axillary nerve

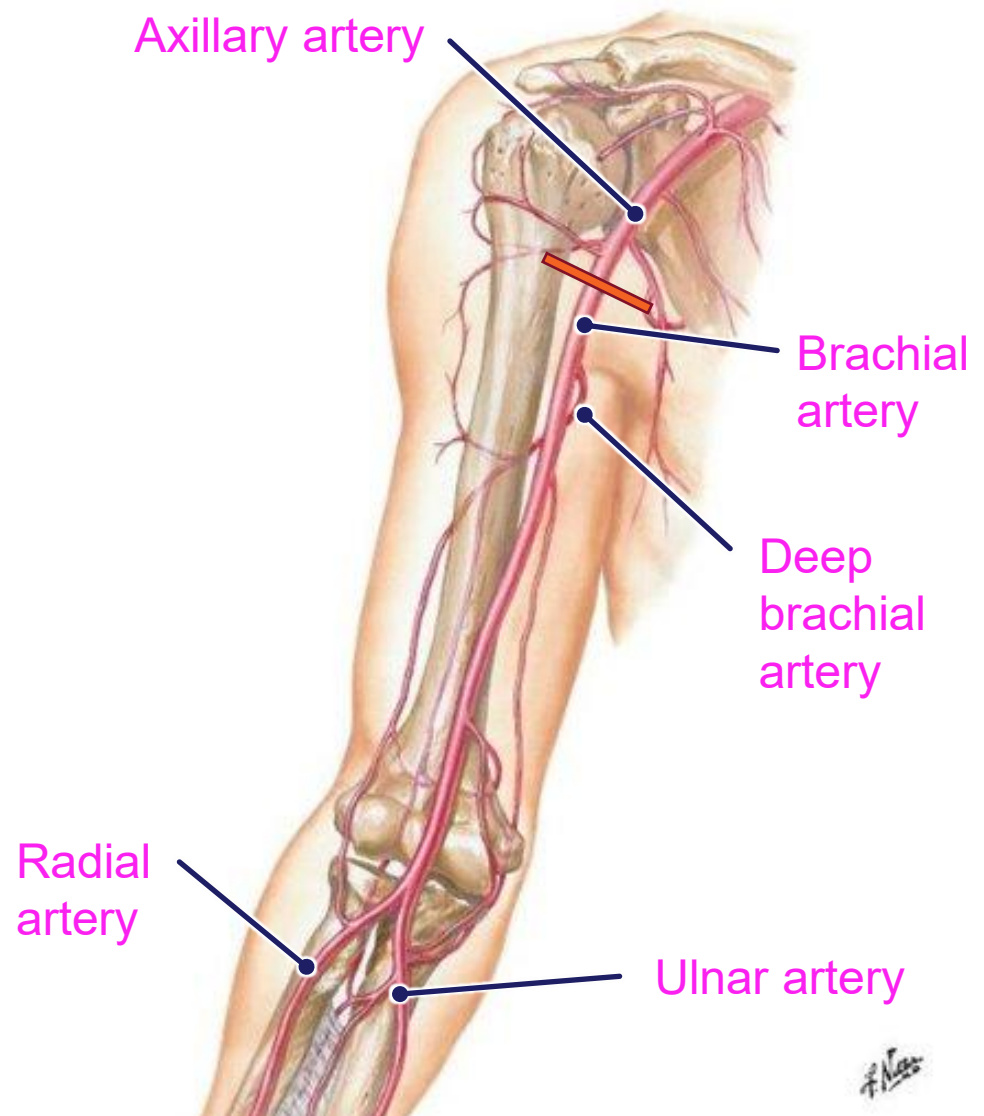
- Runs with **posterior humeral circumflex artery** in the quadrangular space

Radial nerve

- Runs with the deep **brachial artery** in the **radial groove**

Blood Supply of the Arm

- Axillary artery becomes the brachial artery at the inferior border of the **teres major**
- Deep brachial runs posterior to humerus in the **radial groove**
- Brachial bifurcates into ulnar and radial at or distal to the **elbow**



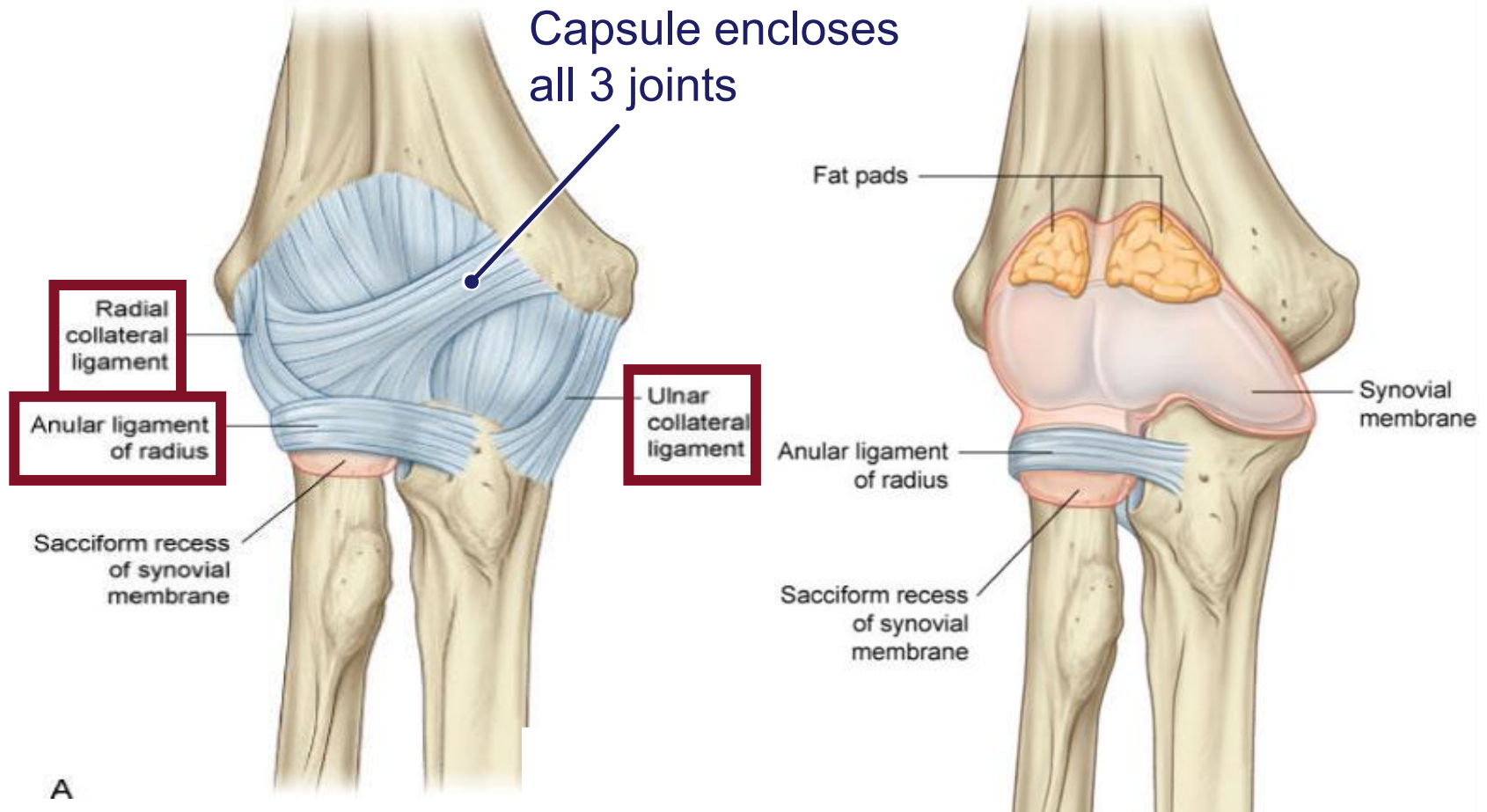
Clinical Correlate: Volkmann's ischemic contracture

- Consequence of **supracondylar fracture** if not recognized and quickly repaired
- Injury to brachial artery can lead to ischemic flexor contracture of the muscles of the anterior compartment



Elbow overview - ligaments

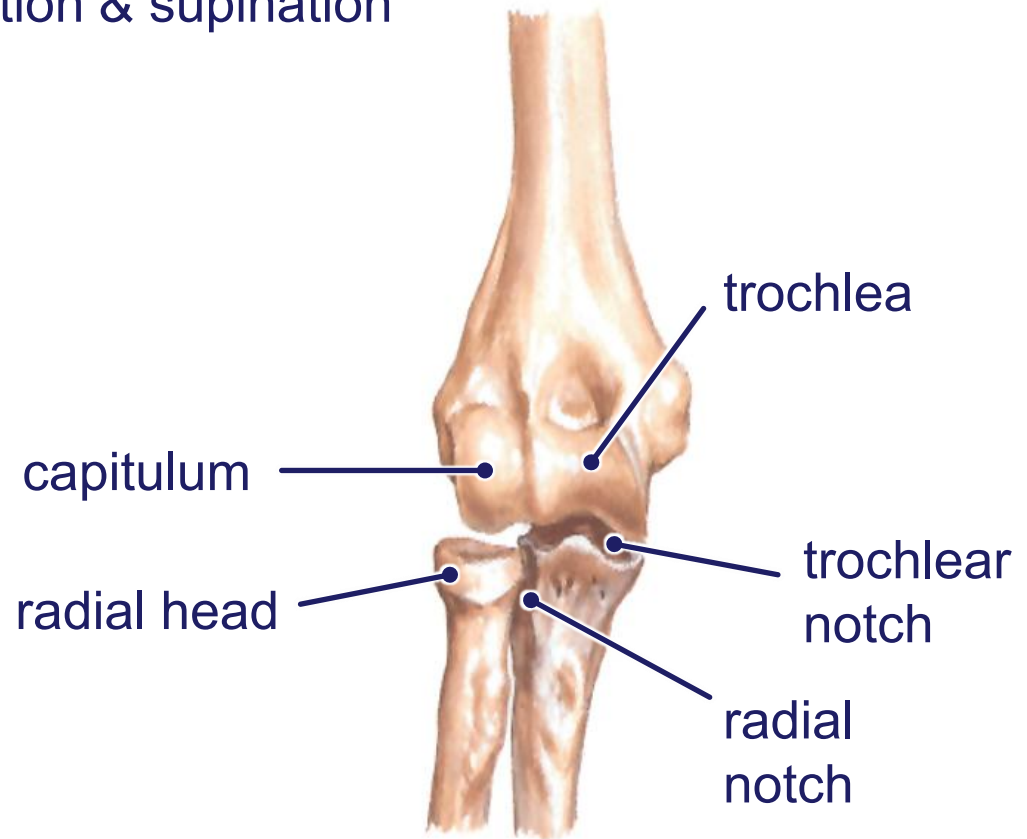
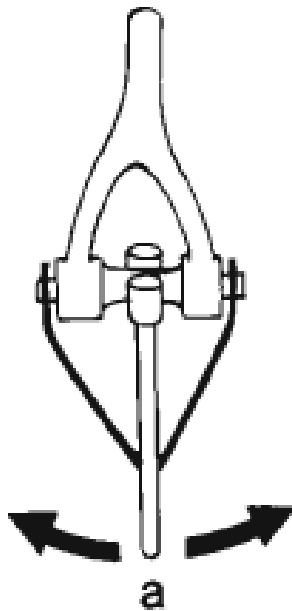
- The elbow joint is a complex joint involve three separate articulations that share a common synovial cavity



Elbow overview - joints

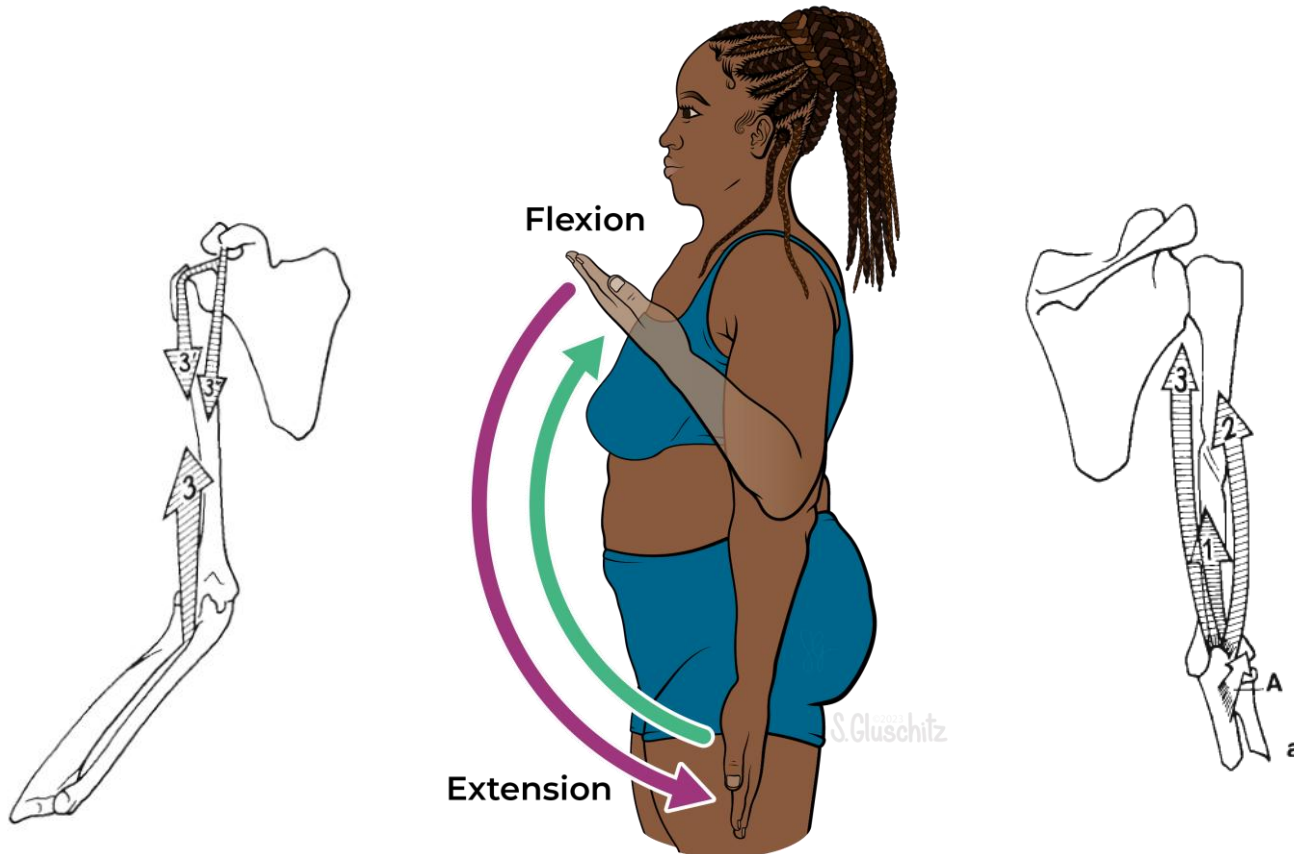
The elbow joint's three separate articulations are:

1. **Humero-radial joint**: flexion & extension
2. **Humero-ulnar joint**: flexion & extension
3. **Radio-ulnar joint**: pronation & supination



Flexion & Extension

- Humero-radial joint: flexion & extension
- Humero-ulnar joint: flexion & extension



Radio-ulnar joint ligaments

- **Proximal**

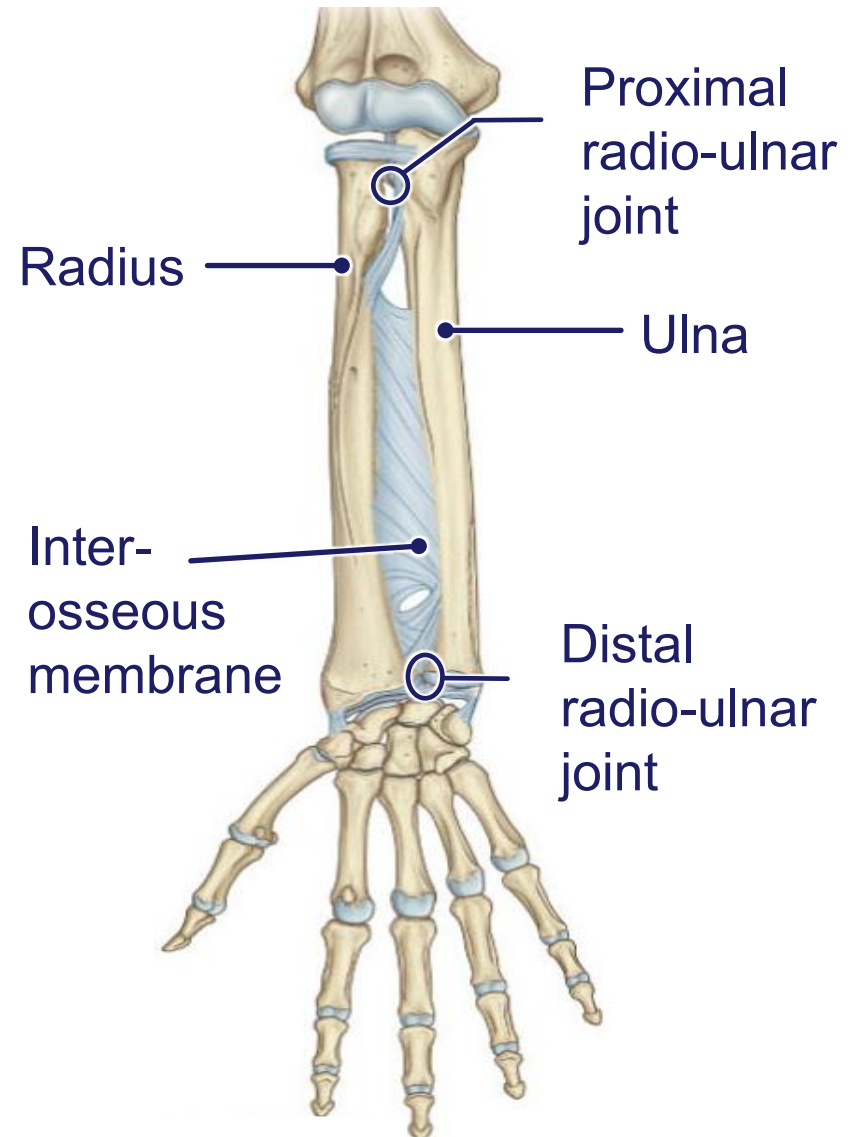
- Radial head with radial notch of ulna
- Held in place by **annular ligament**

- **“Middle”**

- **Interosseous membrane**
 - Sheet of dense CT that runs obliquely between radius and ulna
- Transfers forces from radius to the ulna and then the humerus

- **Distal**

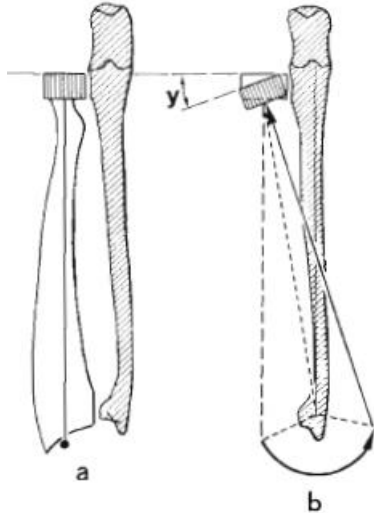
- Articular head of ulna with ulnar notch of radius



Pronation & Supination

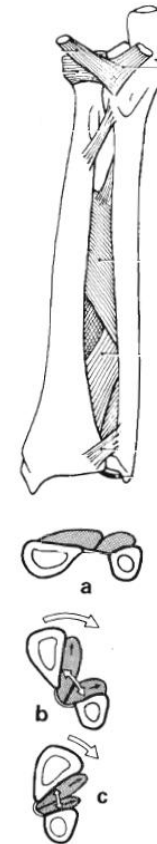
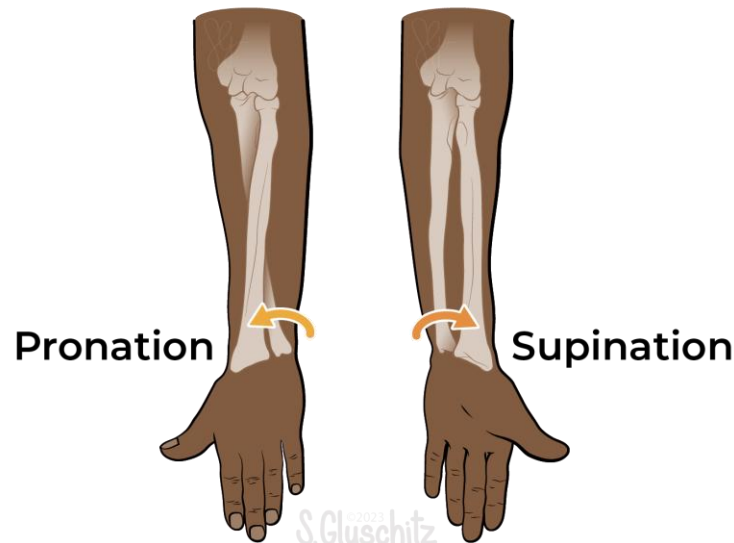
Proximal radio-ulnar joint

- Radial head rotates on the capitulum in the radial notch of the ulna



Distal radio-ulnar joint

- Radius slides anteriorly around the head of the ulna



Clinical Correlate: “Nursemaid’s” (Pulled) Elbow

- Subluxation of the radial head
 - Radial head slips out of the annular ligament
- Commonly occurs in young children (1-4 years old)
- Easily reduced by ‘popping’ the radial head back under the ligament

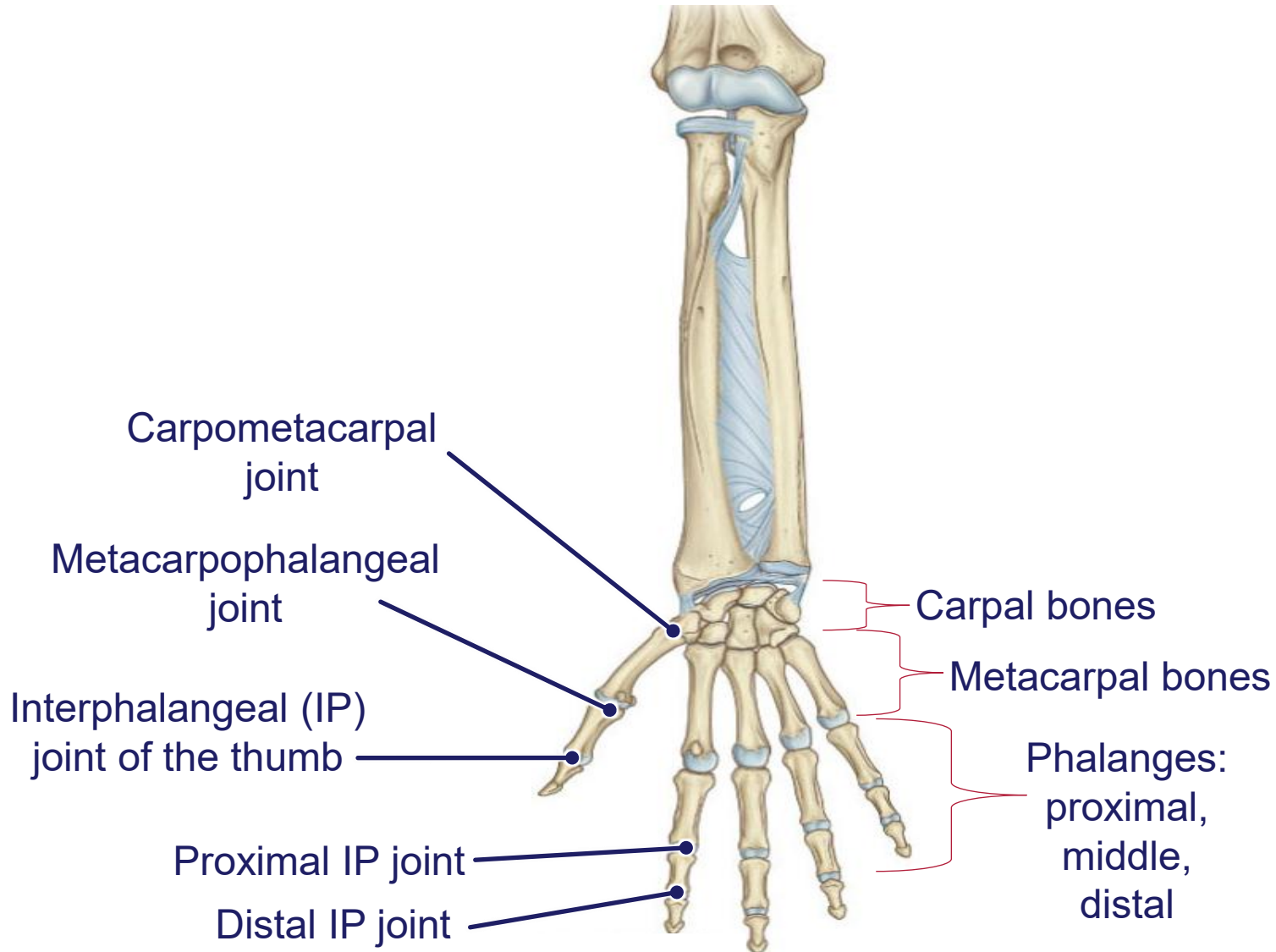


Subluxation



Dislocation

Forearm: Hand Articulations Overview



Forearm: General Layout

- Anterior (flexor) compartment
 - Ulnar nerve (medial one and a half)
 - Median nerve
 - Anterior interosseous (IO) nerve
- Posterior (extensor) compartment
 - Radial nerve
 - Deep radial nerve
 - Posterior interosseous (IO) nerve
- Compartments separated by ***intermuscular septum*** and ***interosseous (IO) membrane***

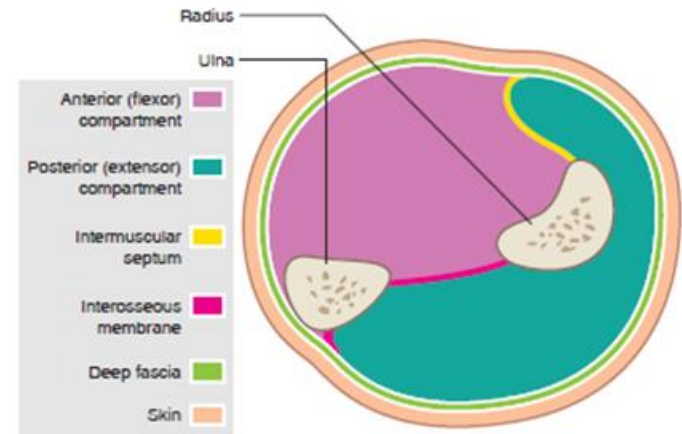
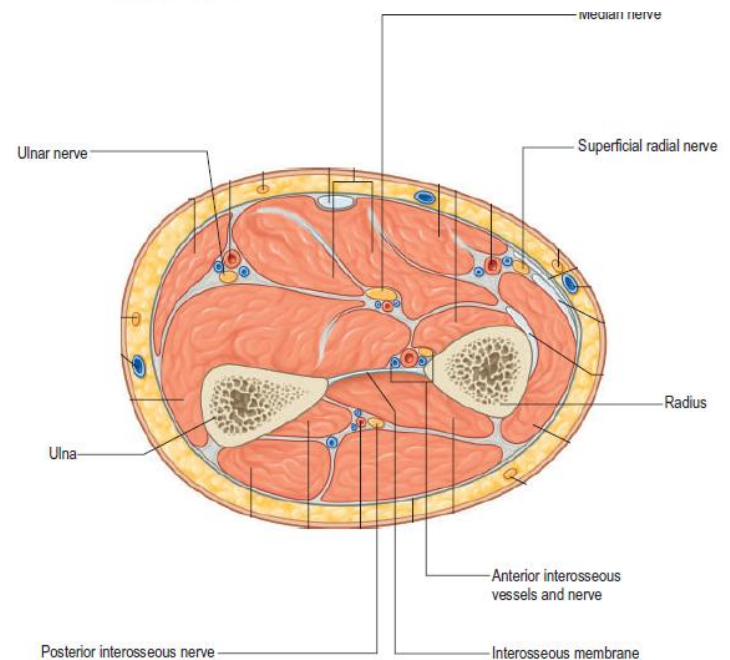


Fig. 15.36



Anterior Compartment – Superficial layer

Flexor carpi radialis

- Flexes and **abducts** the wrist
- **Median nerve**



Pronator teres

- Pronation
- **Median nerve**

Flexor carpi ulnaris

- Flexes and **adducts** the wrist
- **Ulnar nerve**

Palmaris longus

- Flexes the wrist
- Resists shearing forces when gripping
- **Median nerve**

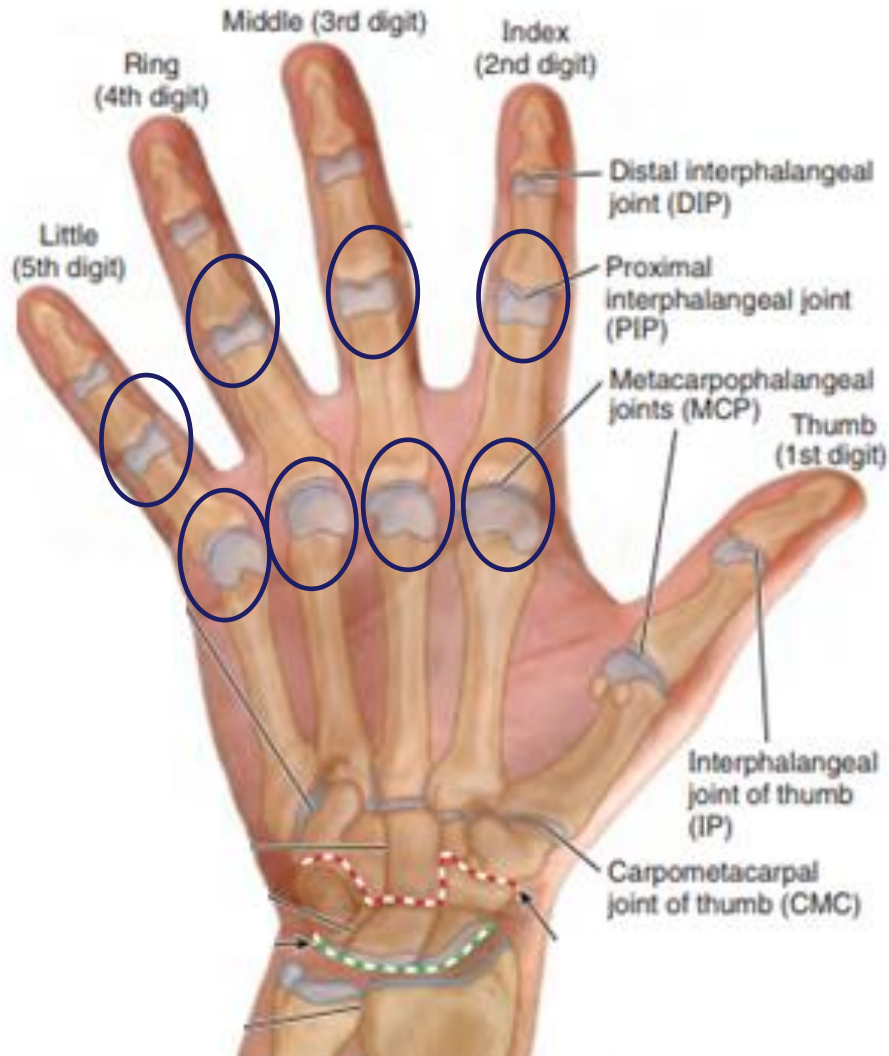
Anterior Compartment – Middle layer

Flexor digitorum superficialis (FDS) —

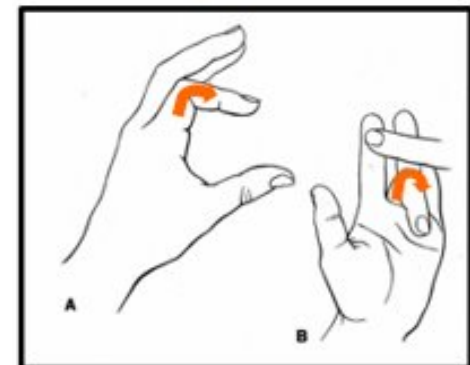
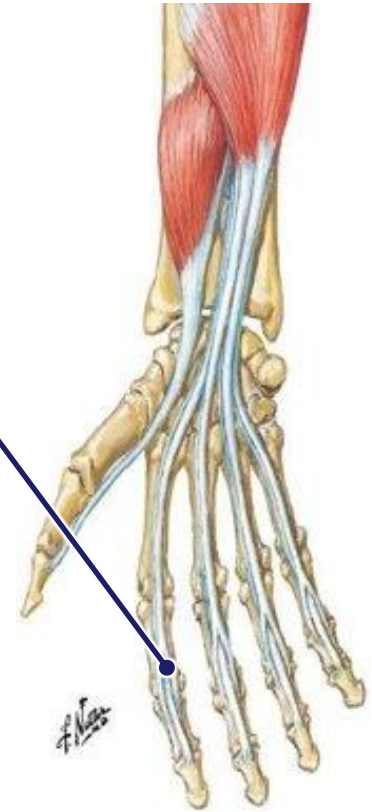
- Flexes the wrist
- Flexes the metacarpophalangeal (MCP) and proximal interphalangeal (PIP) joints of digits 2-5
- Median nerve



FDS movement



FDS
attaches
to the
middle
phalanx



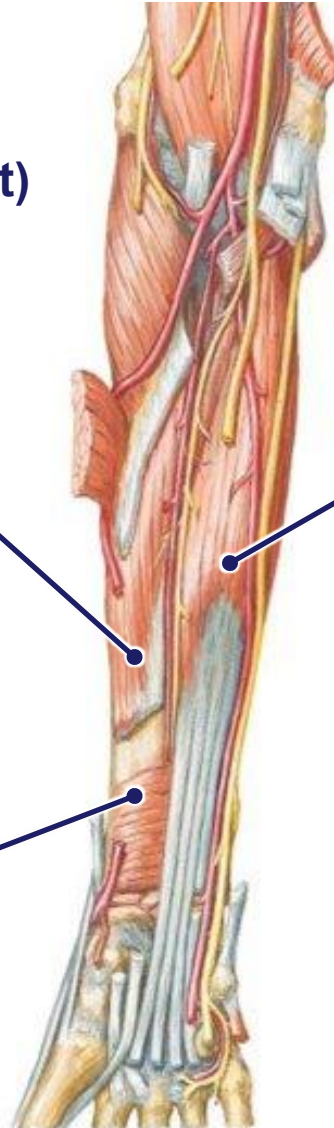
Anterior Compartment – Deep layer

Flexor pollicis longus (cut)

- Flexes MCP and interphalangeal (IP) joint of the thumb
- Anterior IO nerve

Pronator quadratus

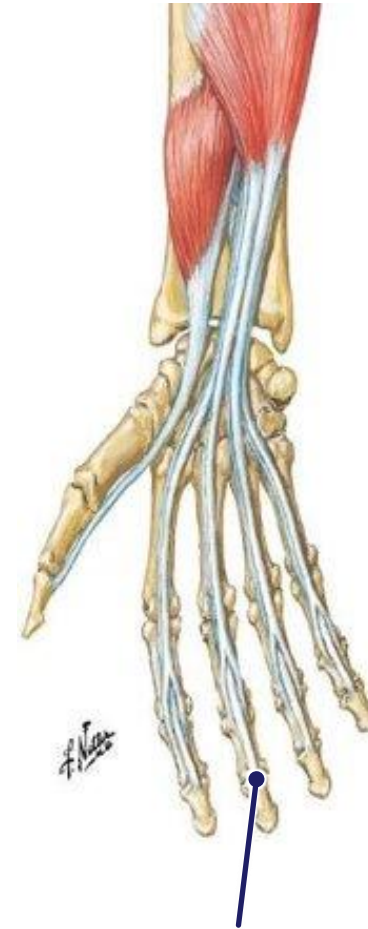
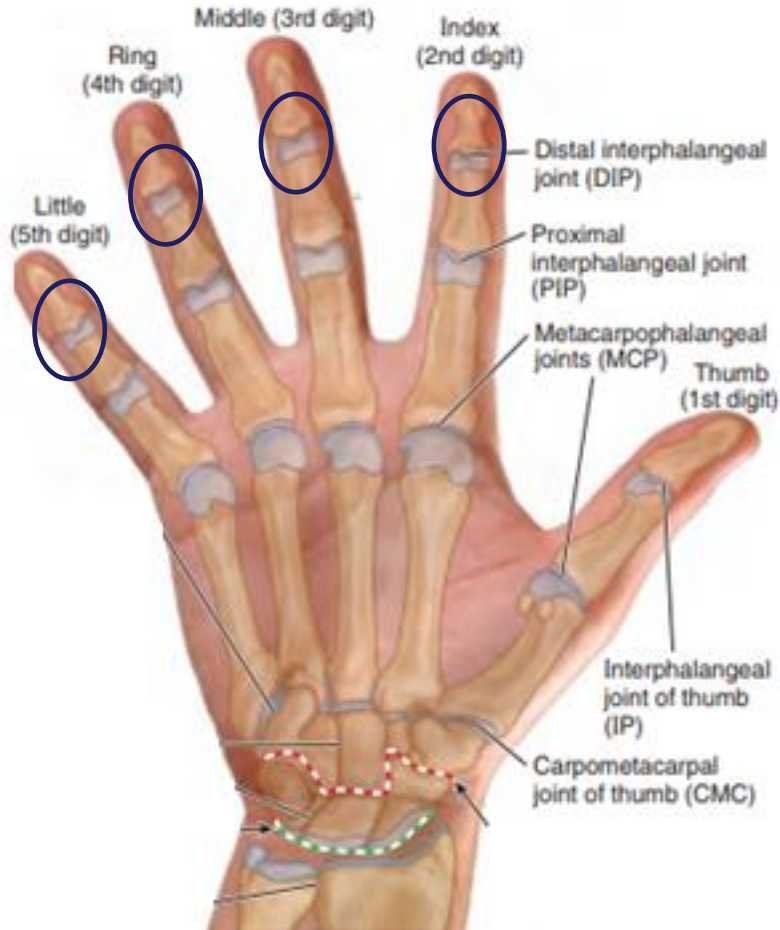
- Pronation of the forearm
- Anterior IO nerve



Flexor digitorum profundus

- Flexes the wrist
- After flexion of MCP and PIP joints, flexes the distal IP (DIP) joints of digits 2-5
- Anterior IO nerve (lateral 2 tendons)
- *Ulnar nerve (medial 2 tendons)*

FDP movement



FDP attaches to the distal phalanx



Neurovascular courses in the arm: median and ulnar nerves

Anterior IO nerve

- Branches from the median nerve in ***pronator teres***
- Lies on anterior surface of IO membrane with anterior IO artery



Median nerve

- Lies anterior to medial supracondylar ridge
- Crosses elbow at cubital fossa
- Pierces ***pronator teres***
- Then runs deep to ***FDS*** down to the wrist

Ulnar nerve

- Crosses elbow posterior to medial epicondyle
- Runs beneath ***flexor carpi ulnaris*** with ulnar artery

Posterior Compartment – Superficial layer

Extensor carpi ulnaris

- Extends and **adducts** the wrist
- **Posterior IO nerve**

Extensor digitorum

- Extends wrist, MCP joints, PIP and DIP joints of digits 2-5
- **Posterior IO nerve**

Extensor digiti minimi

- Extends MCP joints, PIP and DIP joints of the little finger
- **Posterior IO nerve**

Brachioradialis

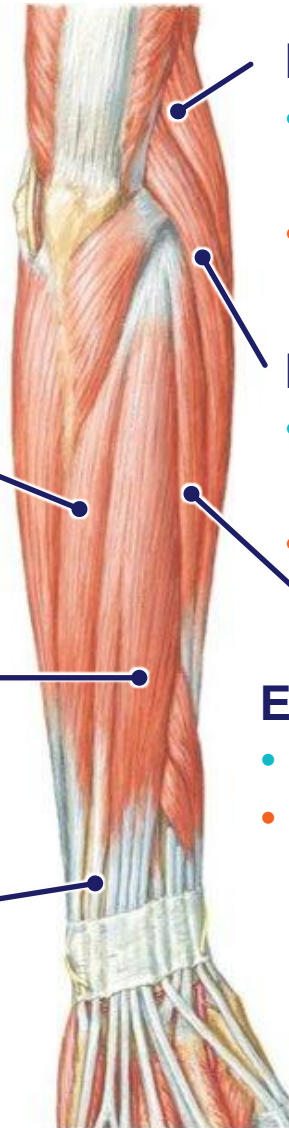
- **Accessory flexor** when the hand is in midprone position
- **Radial nerve**

Extensor carpi radialis longus

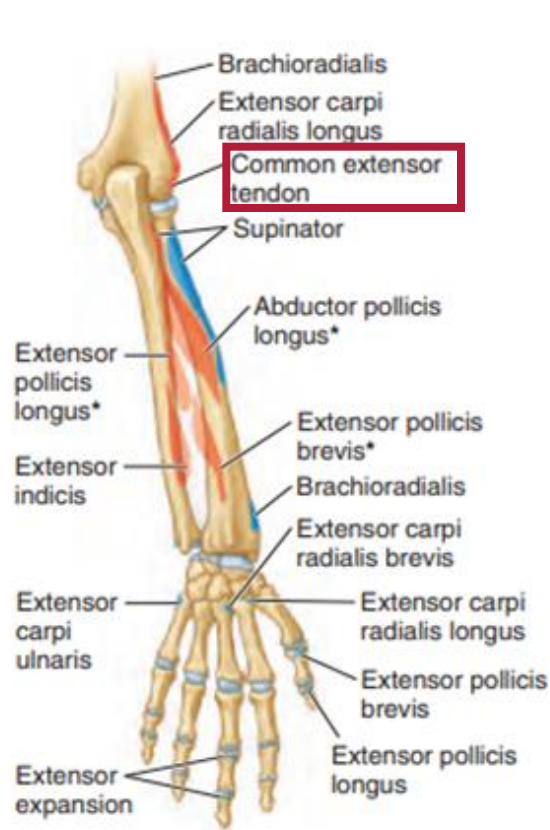
- Extends and **abducts** the wrist
- **Radial nerve**

Extensor carpi radialis brevis

- Extends and **abducts** the wrist
- **Deep branch of the radial nerve**

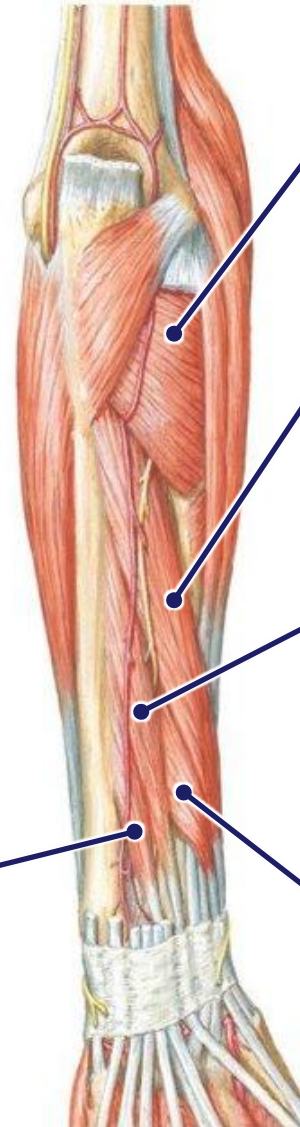


Posterior Compartment – Deep layer



Extensor indicis

- Extends the index finger
- Posterior IO nerve



Supinator

- Supination of forearm
- Posterior IO nerve

Abductor pollicis longus

- Abducts CMC joint of thumb
- Posterior IO nerve

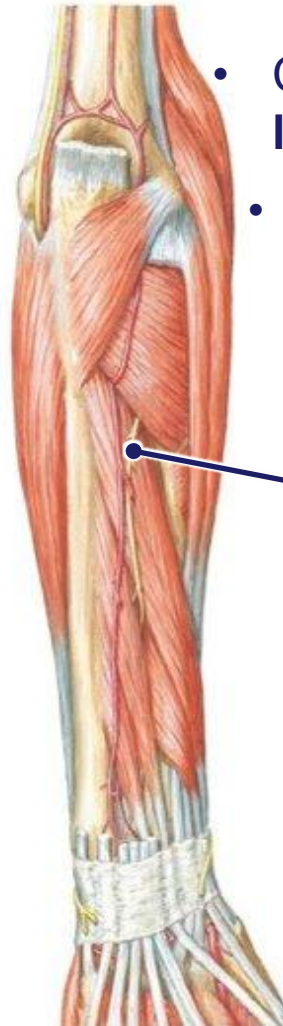
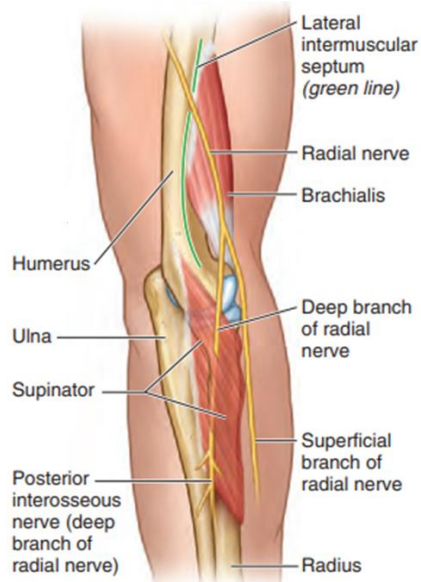
Extensor pollicis longus

- Extends the CMC and IP joint of thumb
- Posterior IO nerve

Extensor pollicis brevis

- Extends CMC joint and MCP joint of thumb
- Posterior IO nerve

Neurovascular courses in the arm: radial nerve



Radial n

- Crosses the elbow anterior to the **lateral epicondyle**
- Divides into superficial and deep

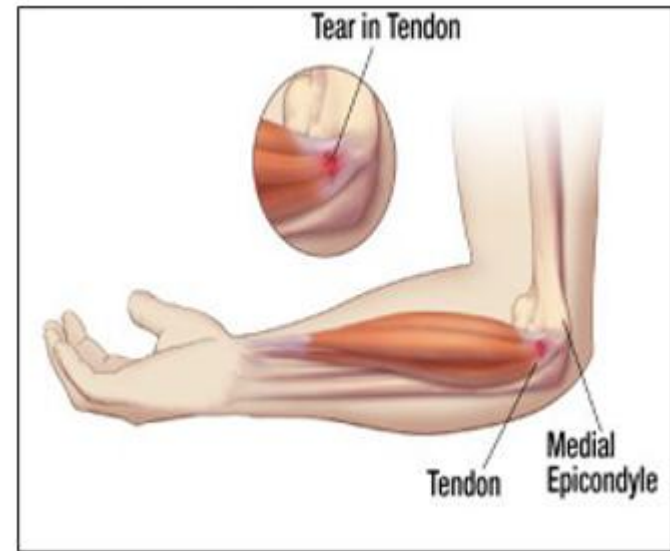
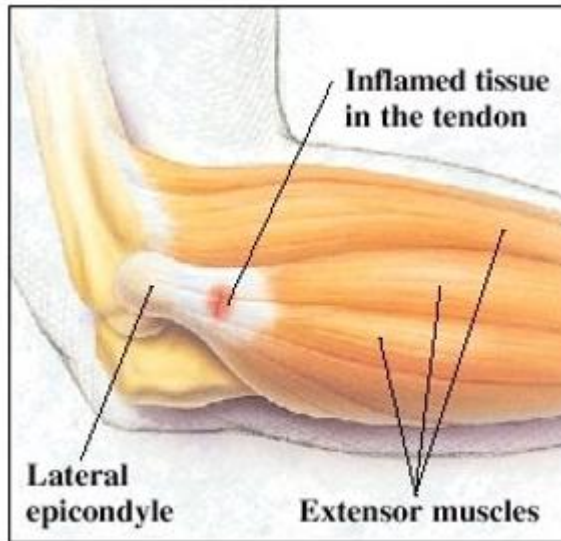
Deep radial pierces **supinator** emerging as the **posterior IO nerve (motor)**

- Superficial radial n (sensory)**
- Runs beneath **brachioradialis** down to the dorsum of the hand



Clinical Correlate: Epicondylitis (inflammation of the common tendons)

- “Tennis elbow”
 - Lateral epicondylitis
 - Common extensor origin
- “Golfer’s elbow”
 - Medial epicondylitis
 - Common flexor origin



Cubital Fossa

• Borders

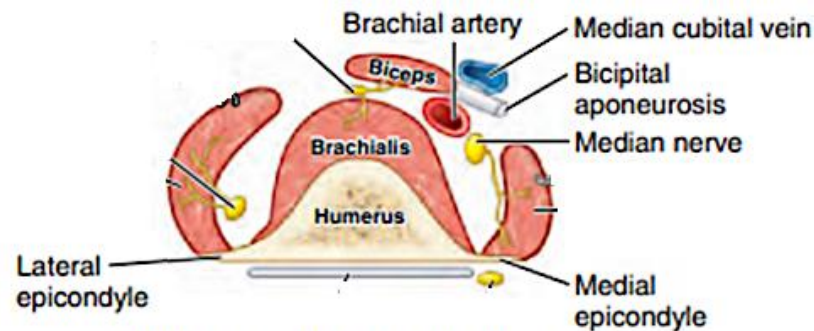
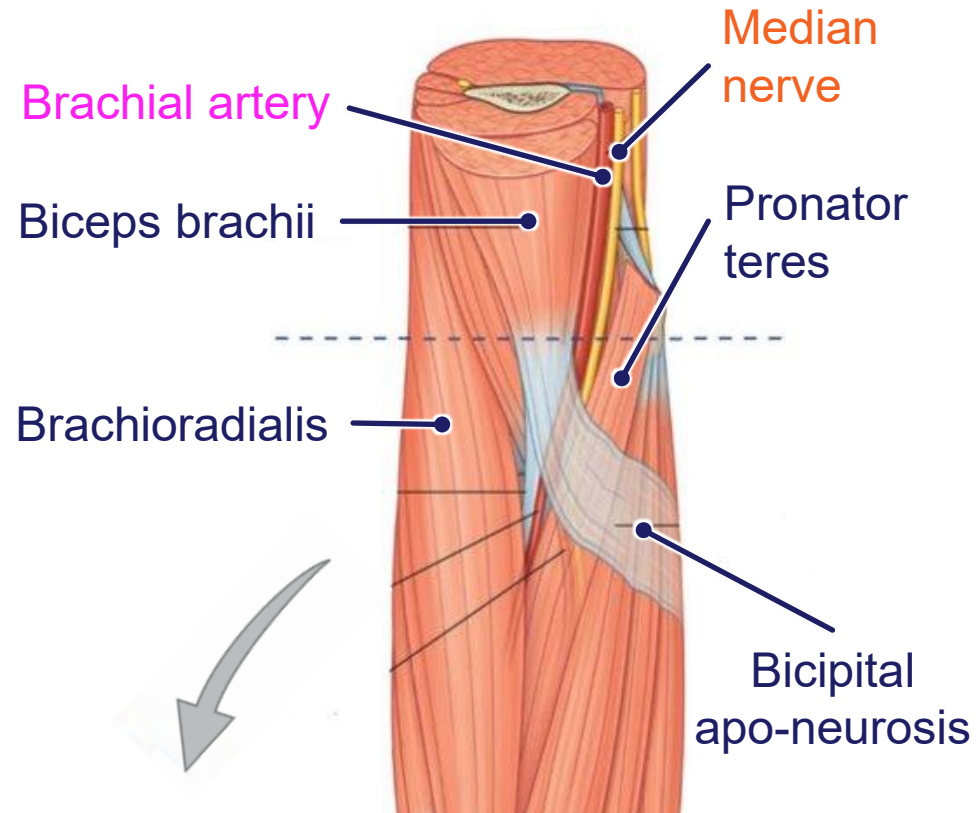
- Line between the **epicondyles**
- Pronator teres (medially)
- Brachioradialis (laterally)

• Contents – medial to lateral

- Median nerve
- Brachial artery
- Biceps tendon

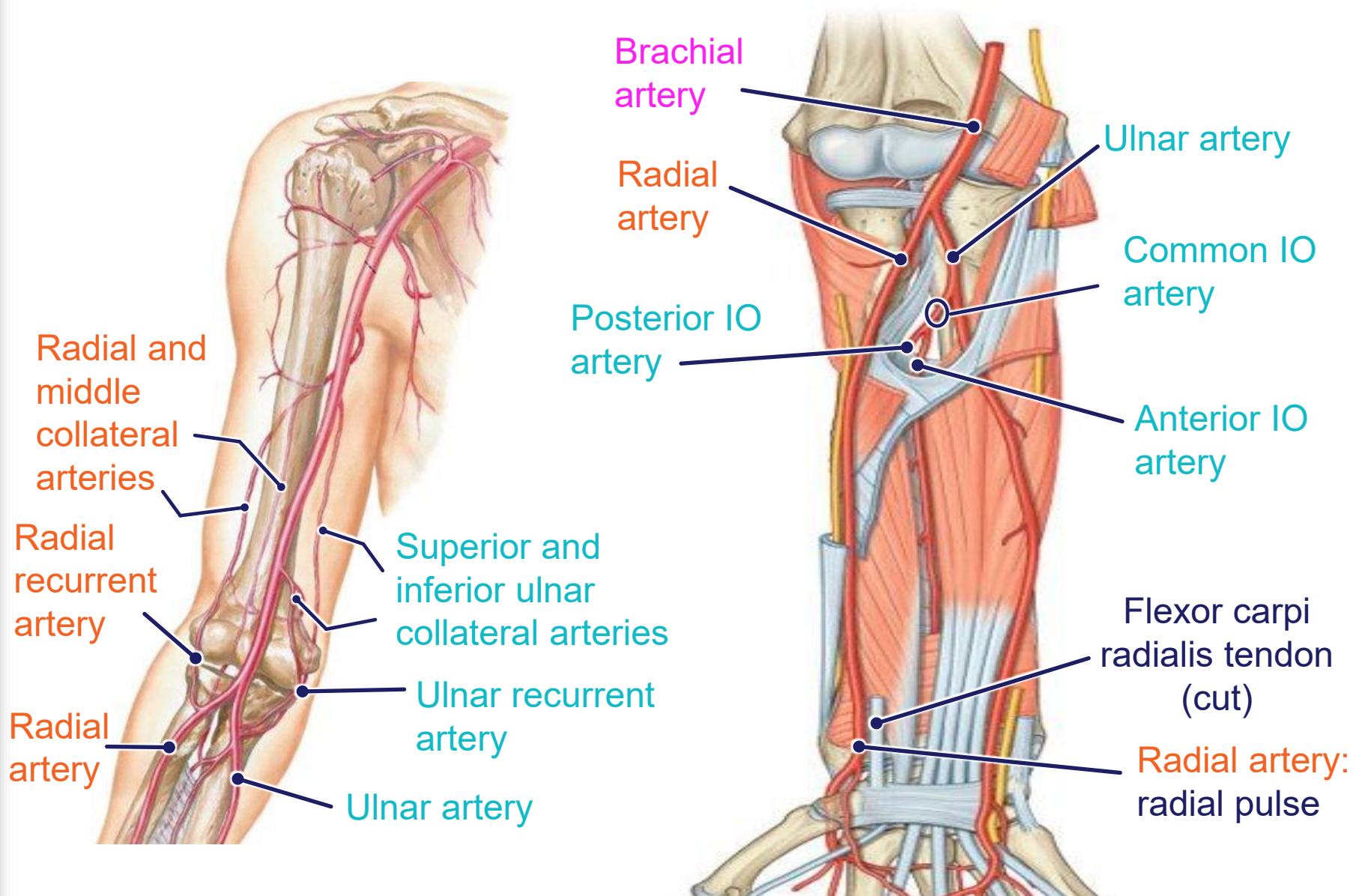
• Floor: brachialis muscle

• Roof: skin, fascia, median cubital vein, bicipital aponeurosis



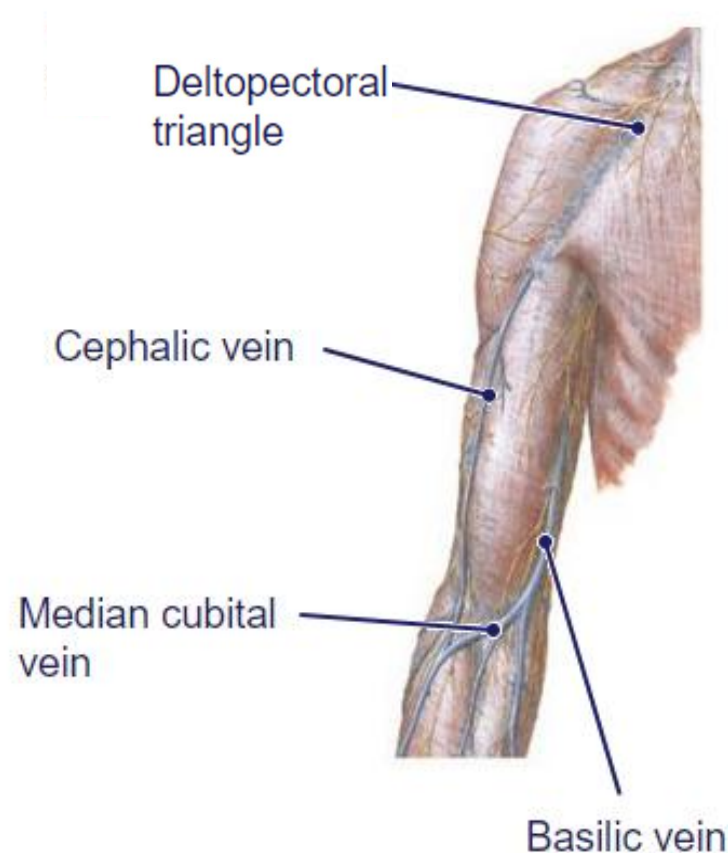
(A) Cross section of cubital fossa

Blood Supply of the Elbow and Forearm



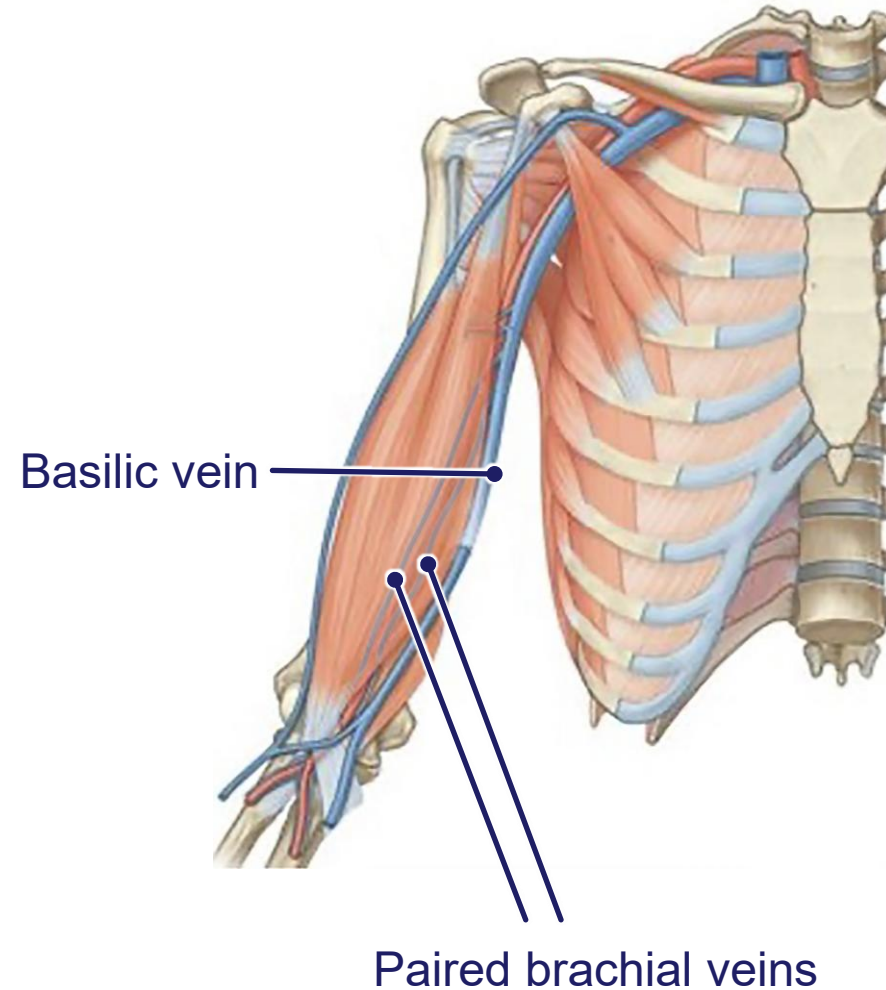
Superficial venous drainage of the arm and forearm

- **Superficial forearm**
 - Basilic vein on the **medial side** of the forearm
 - Cephalic vein on the **lateral side** of the forearm
- **Superficial arm (in skin and fascia)**
 - **Median cubital vein** interconnects basilic and cephalic veins
 - **Basilic vein** eventually becomes deep to form **axillary vein**
 - **Cephalic vein** (lateral) empties into axillary vein at the apex of the axilla

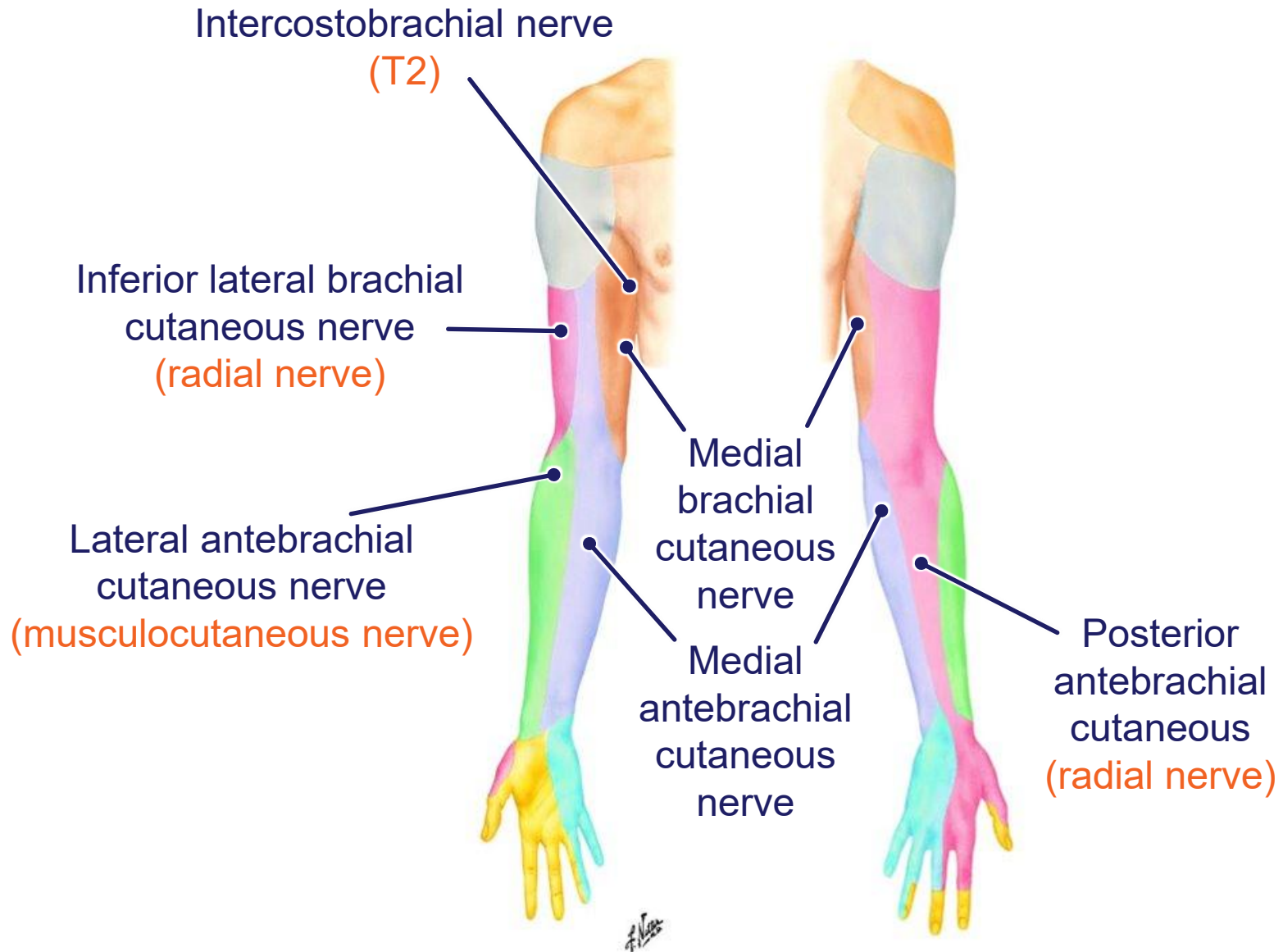


Deep venous drainage of the arm and forearm

- **Deep forearm**
 - **Paired radial and ulnar veins** run with the radial and ulnar arteries in their compartments
- **Deep arm (in the compartment)**
 - **Basilic vein** becomes deep and is joined by paired brachial veins
 - Becomes **axillary vein** at the lower major of teres major



Cutaneous Innervation: Arm & Forearm



Complete the table (for self-review)

Movement of the elbow joint	Nerves affected	Muscles affected	Functions lost
Flexion			
Extension			
Pronation			
Supination			

fin